Dear Hiring Committee,

I saw the job post on **LinkedIn** and thought I'd be a good fit for it.

I am a Master's student in Computer Science at the University of Cincinnati, graduating December 2026, looking for a role as an **AI Deep Learning/ML Engineer** at **Mentium Technologies**. With hands on PyTorch, TensorFlow, and OpenCV experience building and evaluating vision model pipelines, plus production software engineering experience, I am excited about the opportunity to contribute to Mentium's deep learning library for vision AI models on your in-memory compute hardware.

Here are reasons that make me a great fit for the role:

- **Vision model understanding:** As a Graduate Researcher at the University of Cincinnati, I built a PyTorch-based evaluation harness benchmarking a GPT-4o plus GroundingDINO (object detection) plus CLIP (Vision Transformer) pipeline against a proposed method across 53 runs on 20+ Android apps, achieving a 25.7x latency reduction while preserving task-completion fidelity, work that required close attention to inference time and model efficiency relevant to real-time edge deployment.

- **Python and numerical computing proficiency:** At Siemens Industry Software, I built modular Python automation (Boto3, object-oriented design) that replaced legacy scripts and cut deployment time by 40%, and I regularly use NumPy and pandas across my ML projects for data processing and model evaluation.

- **Modular development and collaborative coding:** I migrated 40+ production environments to Kubernetes using ArgoCD, Helm, and GitLab CI/CD, collaborating through GitHub and producing clear technical documentation, and served as Quality Coach and SonarQube Specialist, raising automated code coverage from 35% to 78%.

- **Hardware and performance awareness:** My research work involved measuring inference time and latency tradeoffs directly, and my NYC Mobility Forecasting project applied PyTorch and scikit-learn to a 2.3M-row dataset, cutting RMSE 43% over a baseline model, experience that translates well to reasoning about model size and precision tradeoffs for edge deployment.

I am available to work with you after my graduation in December 2026. As a Master's student at the University of Cincinnati and someone who values professionalism and growth, I am enthusiastic about contributing to Mentium's mission of advancing vision AI on ultra-low power hardware.

Please find my CV attached below. I look forward to hearing from you. I would be excited to help build a client-facing deep learning library that brings real-time vision AI to the edge.
Regards,
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